

Replication

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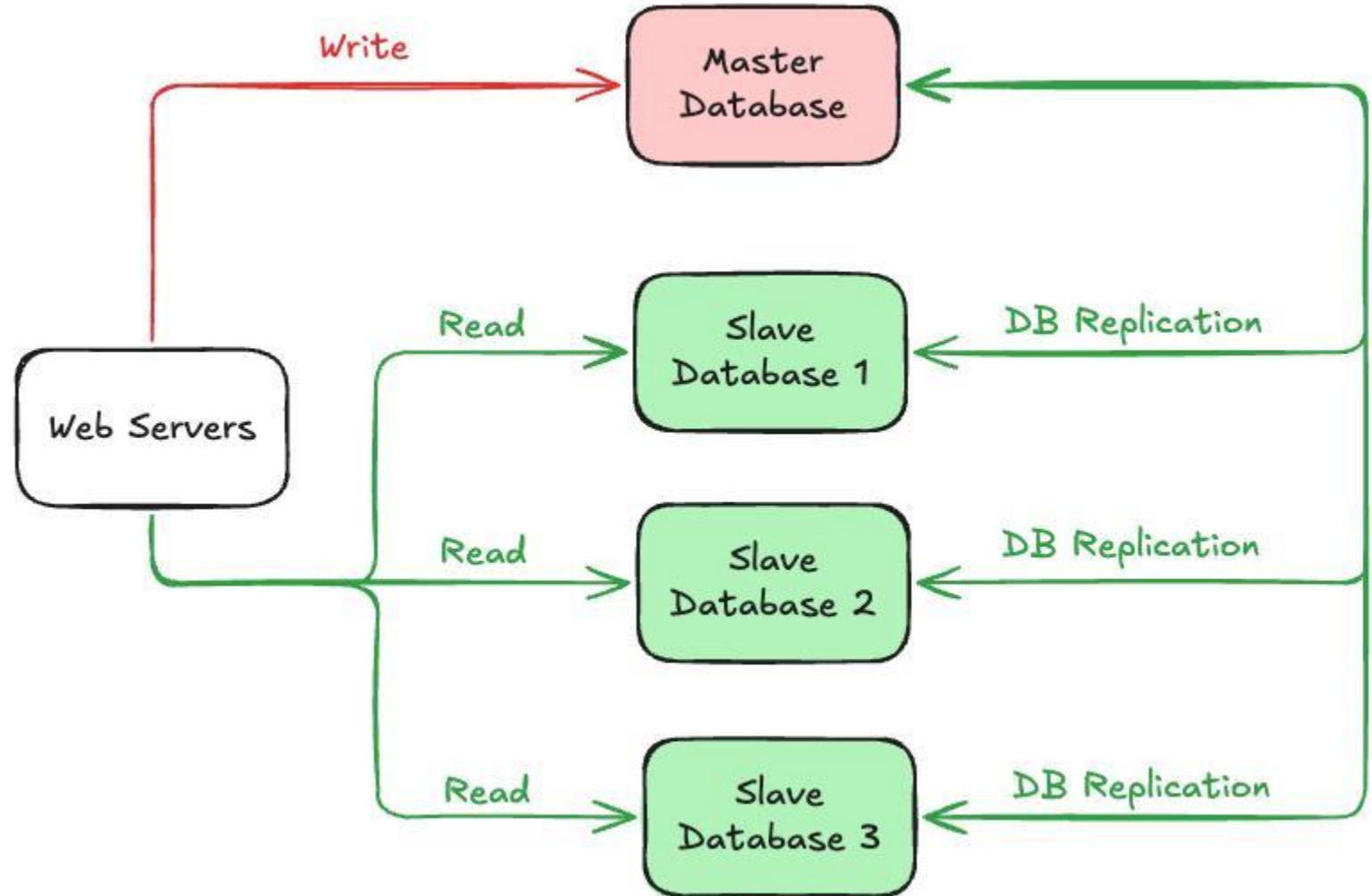
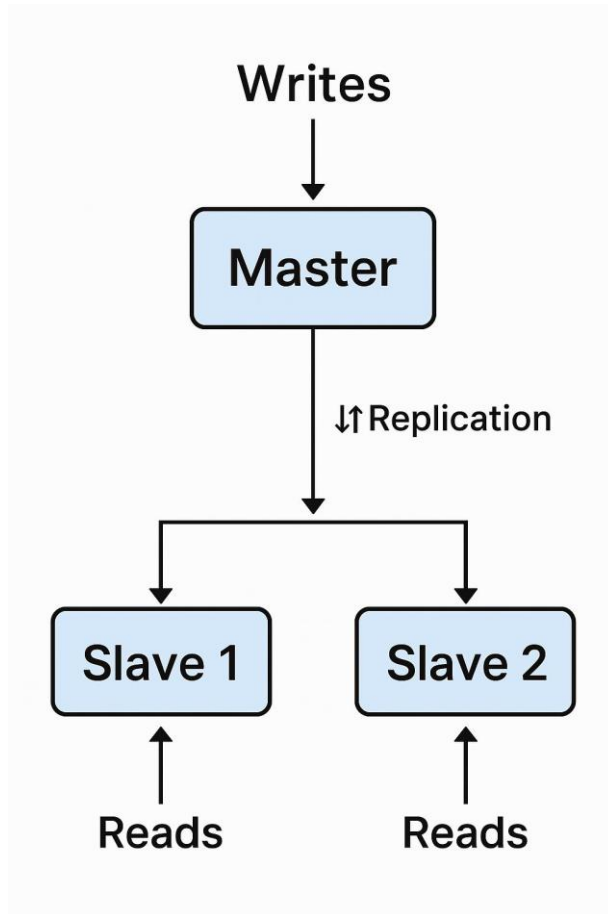
Overview

- **Replication** is the process of **copying data from one database server to another** so that multiple servers contain the same data.
- It improves:
 - Availability
 - Reliability
 - Read performance
 - Backup and disaster recovery

Master- Slave Replication

- In **Master-Slave Replication**, one server acts as the **Master (Primary)** and one or more servers act as **Slaves (Replicas)**.
- **Master**
 - Accepts all write operations (INSERT, UPDATE, DELETE)
 - Sends changes to slaves
- **Slave**
 - Receives copied data from the master
 - Usually handles only read operations

Master- Slave Replication



Advantages and Disadvantages

- **Advantages**

- Fast read performance
- Backup available
- Easy scaling for reads

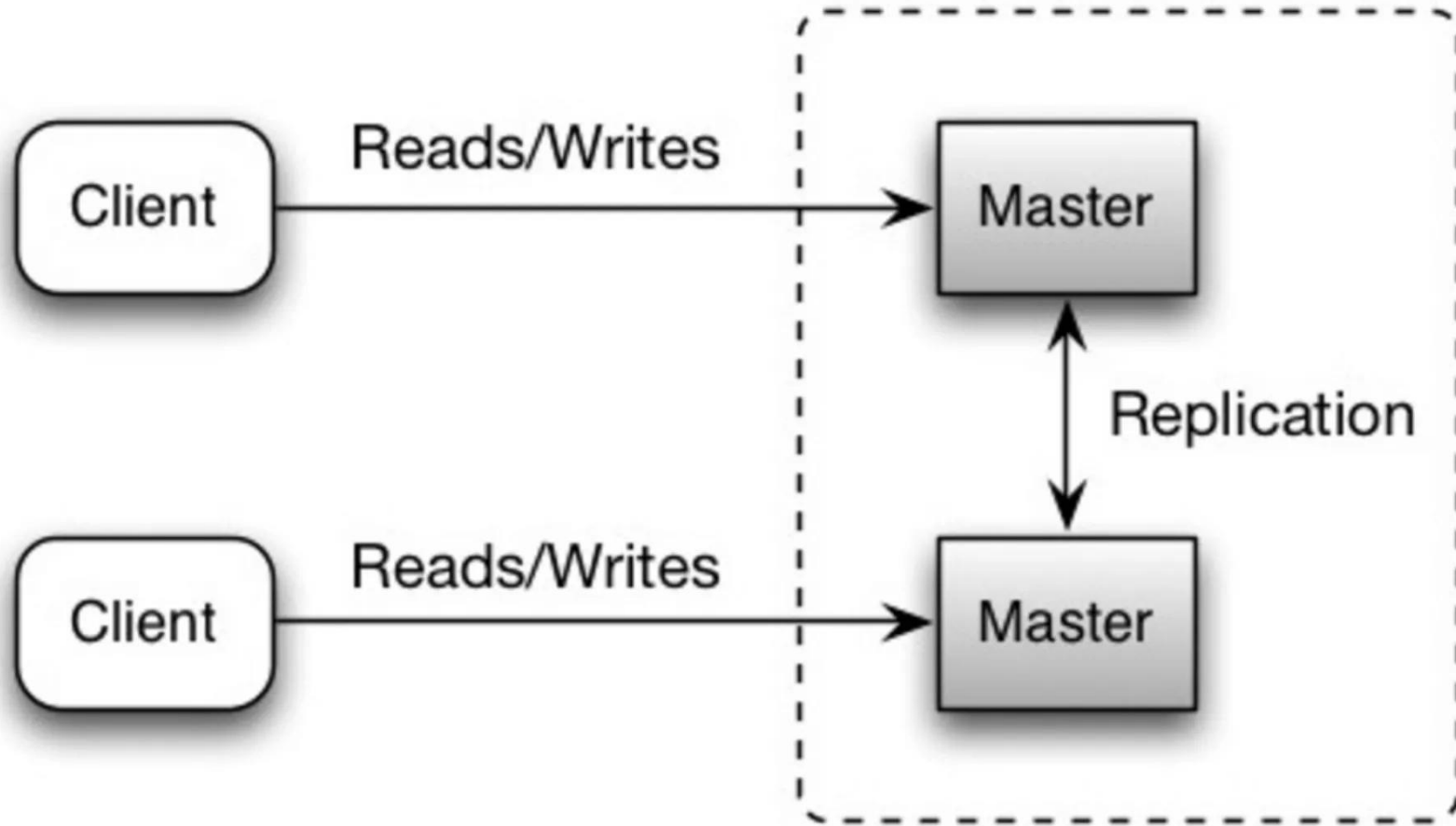
- **Disadvantages**

- Master is a single point of failure (unless failover exists)
- Writes cannot be distributed

Master-Master Replication

- In **Master-Master Replication**, both servers are masters.
- Both databases can:
 - Read
 - Write
 - Replicate to each other

Master-Master Replication



Example

- Two offices:
 - Delhi Office
 - Mumbai Office
- Delhi employee inserts
 - `INSERT INTO Employee VALUES(1,'Amit');`
- The record is copied to Mumbai.
- Mumbai employee inserts
 - `INSERT INTO Employee VALUES(2,'Riya');`
- The record is copied back to Delhi.
- Now both databases contain same data

ID	Name
1	Amit
2	Riya

Conflict Problem

- Suppose both databases update the same row simultaneously.
- Delhi:
 - Salary = 60000
- Mumbai:
 - Salary = 65000
- Now conflict resolution is needed.
- Possible methods:
 - Latest timestamp wins
 - Priority-based master
 - Manual resolution

Advantages and Disadvantages

- **Advantages**

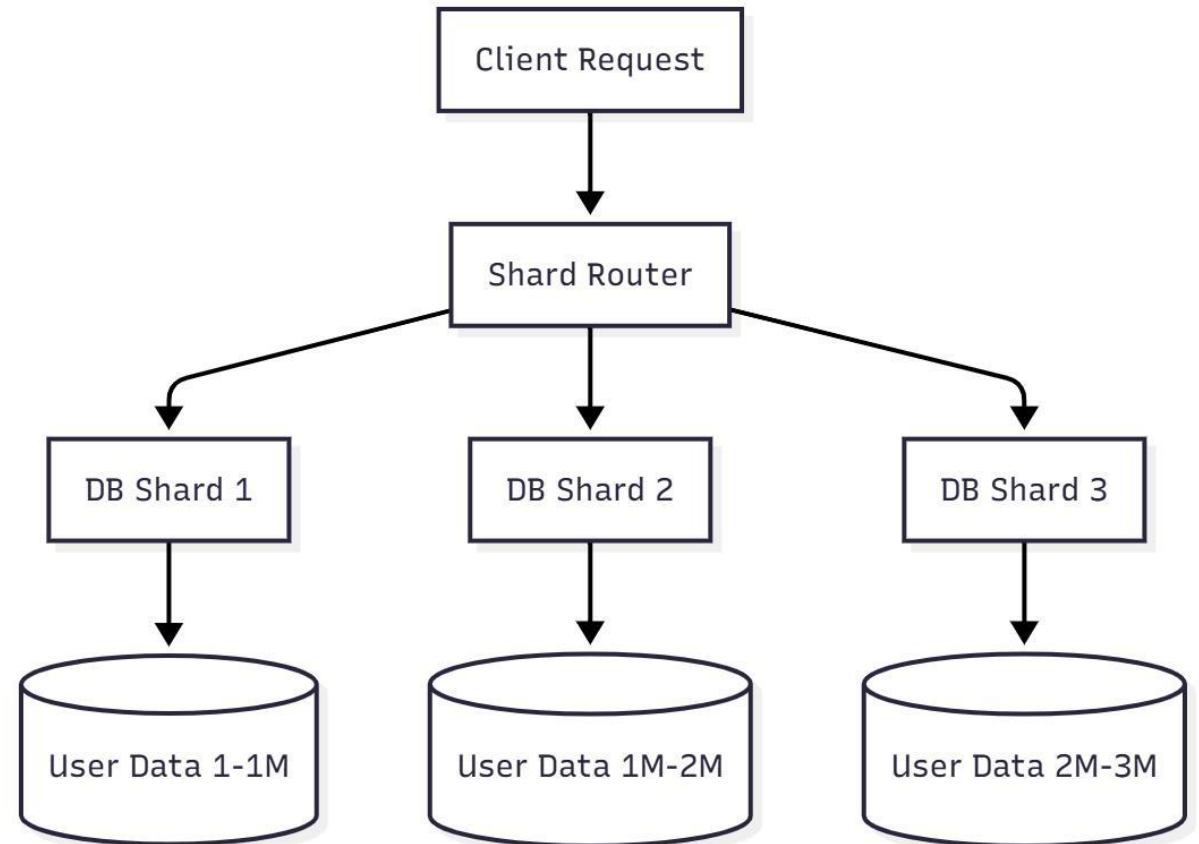
- High availability
- No single point of failure
- Faster writes from multiple locations

- **Disadvantages**

- Conflict handling
- More complex setup

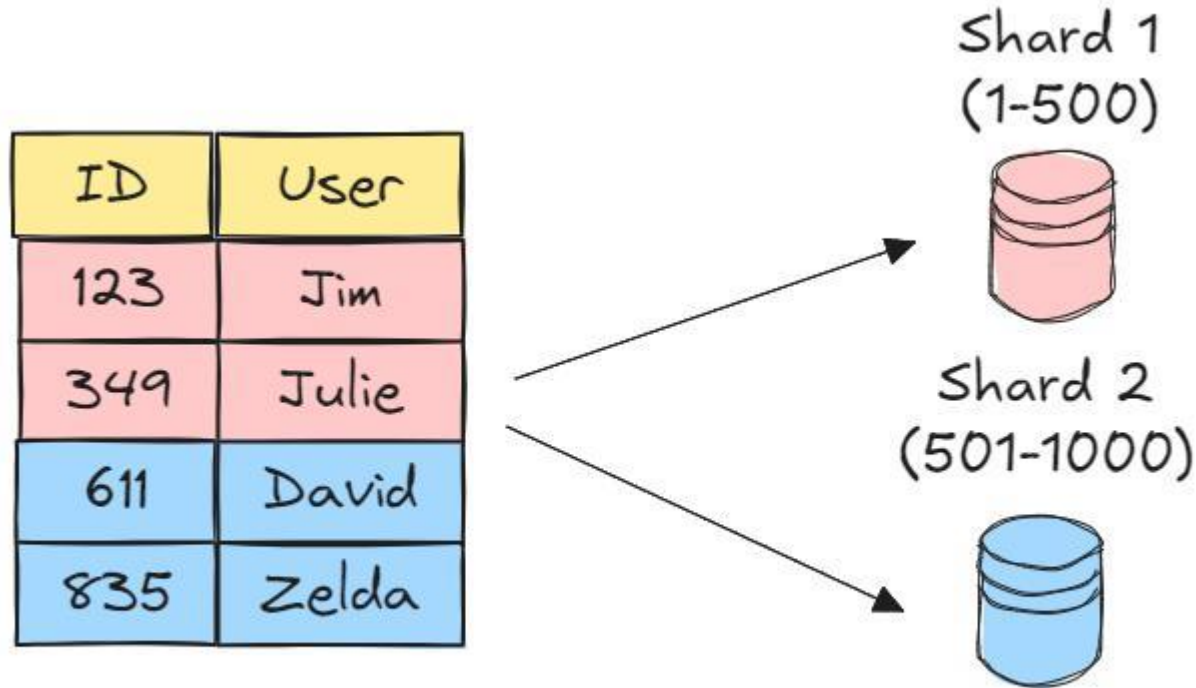
Sharding

- **Sharding** means **dividing** a **very large database** into **smaller databases** called **shards**.
- Each shard stores only part of the data.
- Instead of copying data (replication), sharding **splits** data.



Horizontal Partitioning (Horizontal Sharding)

- Rows are distributed among multiple databases.

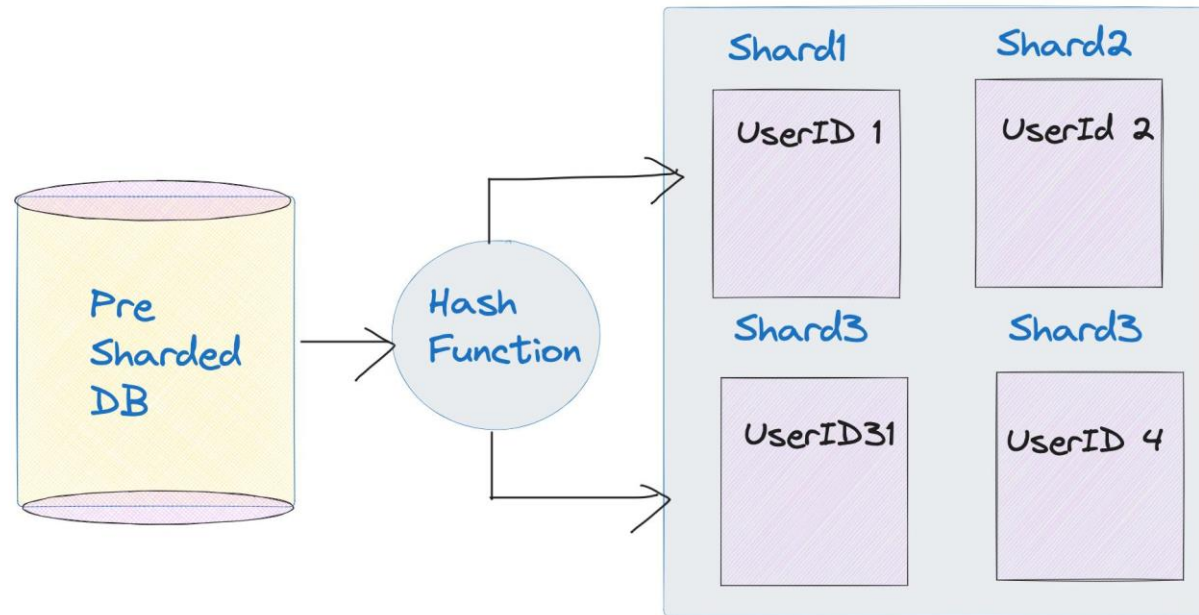


Advantages

- Faster queries
- Better scalability
- Lower storage per server

Sharding Keys

- A **Sharding Key** determines **which shard stores a record**.
- Think of it as an address.



Example 1 : UserID

- Suppose
 - $\text{Shard} = \text{UserID} \% 3$

User ID	Shard
1	1
2	2
3	0
4	1
5	2
6	0

$\text{UserID} = 25$

$25 \% 3 = 1$

Stored in Shard 1

Example 2: Country

India → Shard 1

USA → Shard 2

Japan → Shard 3

- **Good Sharding Key**

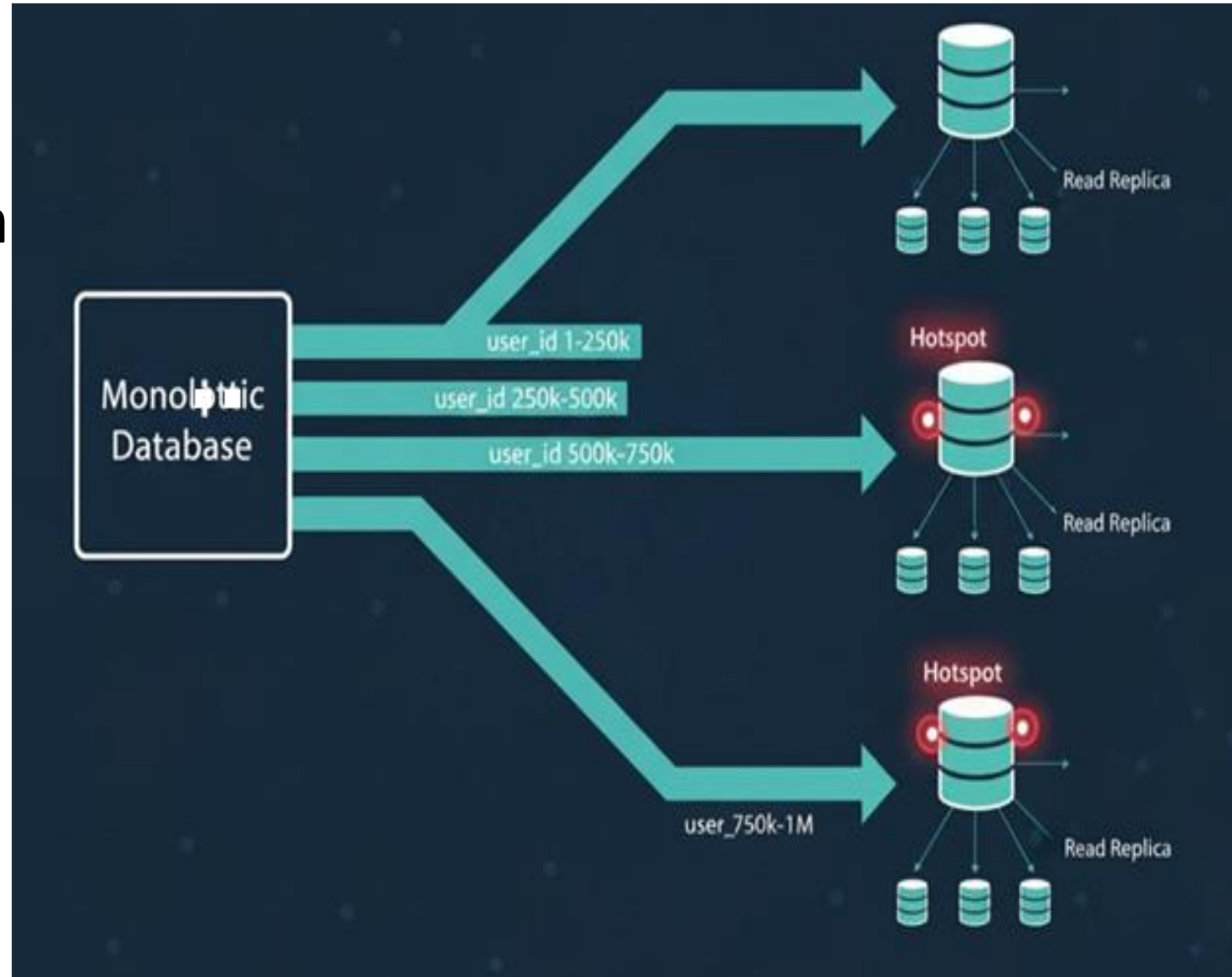
- A good key should:
 - Distribute data evenly
 - Avoid overload
 - Support fast lookup
 - Rarely change
- Examples
 - Customer ID
 - Order ID
 - User ID

- **Bad Sharding Key**

- Example
 - Gender
- Only
 - Male
 - Female
- One shard may contain almost all records.

Hotspot Problem

- A **Hotspot** occurs when one shard receives far more traffic than the others.
- This creates an imbalance.



Example

- Using
 - Date
- All today's orders
 - 2026-06-26
- go to one shard.
- That shard becomes **overloaded** while older shards remain idle.

Problems and Solutions

• Problems Caused

- Slow queries
- Increased latency
- High CPU usage
- Server crashes
- Uneven resource utilization

• Solutions

- Use **hash-based sharding** to spread data more evenly.
- Choose a **high-cardinality sharding key** (e.g., User ID instead of Country).
- Split overloaded shards into smaller shards (**resharding**).
- Use **load balancing** and **caching** for frequently accessed data.

Replication vs Sharding

Feature	Replication	Sharding
Purpose	Copy data	Split data
Data	Same on all replicas	Different on each shard
Improves	Availability and read performance	Scalability and storage
Writes	Usually one master (or coordinated in multi-master)	Distributed across shards
Storage	More total storage due to copies	Storage spread across servers
Example	One customer record exists on every replica	Customer records are distributed across different shards

Thank You